

Introduction to the Special Issue on Recommender Systems for Good

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In this special issue, we bring together contributions that frame recommender systems for good as a broader reorientation of recommender systems research toward societal impact, responsibility, inclusion, and beneficial intervention. The papers collected here help define the emerging RS4Good agenda, examine its risks and normative constraints, study high-stakes and vulnerable contexts, and demonstrate constructive applications in domains such as healthcare, education, tourism, and sustainability. Taken together, they show that recommender systems for good are defined not only by the domains in which they are deployed, but also by the values, design processes, and evaluation criteria that guide their development.

CCS Concepts: • **Information systems** → *Recommender systems; Personalization*; Collaborative and social computing systems and tools; • **Human-centered computing** → *Human computer interaction (HCI)*;

Additional Key Words and Phrases: Recommender systems, social impact, human-centered design, responsible AI, fairness, explainability, accountability

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1 Introduction

Recommender systems research has long been driven by a relatively narrow set of technical and commercial objectives, including predictive performance, ranking quality, engagement, and conversion [12]. While these goals have enabled substantial methodological progress, they have also

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privileged evaluation criteria that leave broader social consequences underexplored. As recommender systems increasingly shape cultural exposure, public discourse, everyday decision-making, and long-term opportunities, questions of who benefits, who is harmed, and according to which values systems are designed have become central [1].

This special issue reflects a growing recognition that recommender systems research must expand beyond narrow optimization objectives toward more explicitly societal concerns. In this sense, recommender systems for good (RS4Good) should not be understood simply as the application of existing techniques to benevolent domains. Rather, as argued by the agenda-setting contributions in this issue, RS4Good entails a broader reorientation of the field: one that foregrounds meaningful impact, interdisciplinary collaboration, human-centered and stakeholder-centered design, as well as forms of evaluation that take long-term outcomes and lived experience seriously [4, 5]. This shift also calls for renewed attention to harms, accountability, justice, inclusion, and explainability.

The papers collected in this special issue make this reorientation visible from several complementary angles. Some articulate the conceptual and methodological foundations of RS4Good; others examine risks, normative constraints, and governance challenges; a further group studies high-stakes contexts in which bias, stereotyping, exclusion, and public responsibility become especially salient; and a final set demonstrates constructive interventions in domains such as healthcare, education, tourism, and sustainability.

2 Framing the Field: RS4Good as a Reorientation of Recommender Systems Research

The first cluster defines what recommender systems for good are and what changes this agenda requires of the field. Its shared premise is that RS4Good cannot be reduced to socially desirable application domains. The notion of “good” implies a broader reorientation of recommender systems research, a rethinking of system goals, represented stakeholders, and evaluation methods. In this sense, RS4Good concerns not only *where* recommenders are deployed, but also *how* they are conceived, designed, evaluated, and governed.

Jannach et al. [5] articulate this reorientation at the level of the field as a whole. Their article surveys use cases in which recommender systems can contribute to societal benefit and argues that progress in recommender systems research should no longer be equated primarily with incremental improvements in predictive performance on narrow benchmark tasks. Instead, they call for stronger attention to socially valuable application scenarios, interdisciplinary collaboration, and evaluation settings that include humans in the loop and attend to long-term consequences. Their contribution provides a conceptual vocabulary for understanding RS4Good as a shift from performance-centered research toward impact-centered research.

Ekstrand et al. [4] complement this field-level agenda by asking how systems for social good should be designed in the first place. They argue that recommender systems intended to promote social good should be designed *with* stakeholders rather than merely *for* them. This moves the discussion beyond socially desirable objectives toward the politics of design: who defines a system’s goals, whose interests are included or excluded, and which forms of knowledge count in the design process. By emphasizing co-design, participation, and democratic accountability, the article shows that RS4Good is not simply a technical agenda with better intentions, but also a methodological commitment to more inclusive and reflexive forms of system development.

Together, these two papers establish the conceptual foundation of the special issue by framing RS4Good around impact, responsibility, and stakeholder engagement.

3 Responsibility Before Intervention: Risks, Justice, Accountability, and Explainability

The second cluster makes clear that socially beneficial recommendation cannot be defined by benevolent intent alone. Systems introduced in the name of relevance, diversity, efficiency, or

even social benefit may still reproduce harms, obscure asymmetries of power, or operate without meaningful accountability. A recommender system can therefore be “for good” only if its design and deployment are accompanied by attention to risks, normative commitments, forms of explanation, and mechanisms of scrutiny.

Kathriarachchi et al. [7] contribute to this discussion through a broad survey of hazards in e-commerce recommender systems and the mitigation strategies proposed to address them. Their contribution is valuable because it demonstrates that questions of social good are not limited to explicitly civic or public-interest domains. Even in one of the most established and commercially mature application areas of recommendation, recommender systems raise concerns that affect consumers, businesses, and society more broadly. By consolidating these risks and responses, the analysis highlights that RS4Good must include a sober understanding of how recommender systems may undermine well-being, autonomy, or fair treatment if such risks are left unexamined.

While Kathriarachchi et al. survey harms and mitigation measures in a major application domain, Laugwitz and Schaetz [10] offer a more explicitly normative account of why good intentions alone are insufficient. Their design justice perspective on news recommendation shifts attention from isolated system properties to broader questions of power, expertise, and structural inequality in the design process. In doing so, the authors argue that the evaluation of social benefit must extend beyond whether a system optimizes a desirable metric. What matters equally is whose values shape the problem definition, who bears the risks of design decisions, and whether recommendation processes reinforce or challenge existing social asymmetries.

Bekavac et al. [2] move the conversation from normative framing to the practical question of oversight. Their article on sock-puppet auditing of very large online platforms highlights the difficulty of evaluating personalized recommender systems in environments where data access is constrained, platform behavior is opaque, and systemic risks may be amplified at scale. The contribution is especially important because it treats accountability not as a desirable afterthought but as an operational challenge that requires new methods, tools, and forms of public scrutiny.

Finally, the contribution on ELSA-compliant explainable recommender systems [6] adds another key element to this cluster by emphasizing the role of explainability in aligning recommender systems with ethical, legal, and social expectations. Within this special issue, explainability is best understood not merely as a technical feature, but as part of the broader responsibility infrastructure of RS4Good.

Together, these papers show that RS4Good depends not only on beneficial goals, but also on the conditions that make such claims credible: risk recognition, normative clarity, explainability, and meaningful scrutiny.

4 High-Stakes and Vulnerable Contexts: Exposure, Inclusion, and Public Responsibility

The third cluster focuses on contexts in which recommender systems influence vulnerable users or mediate access to socially sensitive information. In such settings, recommendation is not merely a matter of convenience or preference matching. It shapes exposure, attention, interpretation, and participation in ways that may have lasting consequences. The social stakes of recommender systems therefore become especially visible when such systems affect children, neurodivergent users, or citizens navigating news environments marked by bias, polarization, and unequal visibility.

Ungruh et al. [16] provide a compelling example of why questions of representational harm belong at the center of RS4Good. By investigating stereotype presence in recommendations that may reach children, the authors highlight how recommender systems can reproduce problematic cultural patterns for a particularly vulnerable audience. The contribution is important not only because it documents the persistence of stereotypes across multiple recommendation strategies, but also because it shows that harms in recommender systems may arise through the subtle ways in

which systems shape cultural exposure, normalize social categories, and participate in the formation of beliefs and expectations.

Khan et al. [8] likewise foreground a user group whose needs are often poorly captured by conventional recommender systems design. Their work on users with ADHD demonstrates that recommendation interfaces and personalization mechanisms can interact with cognitive differences in ways that either support or undermine autonomy, self-regulation, and everyday functioning. This contribution moves the discussion of inclusion beyond generic claims about personalization toward a more situated account of what it means to design recommender systems that are responsive to diverse capacities and vulnerabilities.

Two further papers in this cluster explore related concerns in the domain of news recommendation, where recommender systems do not merely personalize content, but also structure access to public discourse. Ruan et al. [13] address media-bias polarization by proposing automated rewriting strategies that aim to reduce the polarizing effects of biased content while maintaining recommendation quality. Their article treats recommendation not simply as the ranking of available items, but as part of a broader information ecosystem in which design choices affect which kinds of discourse are amplified or attenuated.

The article on bisociative pivoting [9] contributes to this same concern from a different angle. Rather than focusing on rewriting biased content, it studies how the design of recommendation trajectories can moderate users' responses to semantic distance in next-item news recommendation. In the context of this special issue, the contribution is valuable because it questions one of the central tensions in RS4Good: how to broaden exposure without undermining user receptivity.

Taken together, these papers show that RS4Good requires designs attentive to the needs of diverse and potentially vulnerable user groups.

5 Constructive Interventions: Recommender Systems for Better Long-Term Outcomes

The final thematic cluster highlights the constructive potential of recommender systems when they are oriented toward meaningful support in consequential domains. These papers move beyond the mitigation of negative outcomes to examine how recommendation can help users and communities make better decisions, navigate complex environments, and pursue socially valuable goals over time. What unites them is a view of recommendation as a form of guidance whose value lies in improving long-term outcomes rather than maximizing immediate engagement.

Solomon et al. [14] exemplify this constructive orientation in the healthcare domain. Their work on recommending surgical procedures shows how recommender systems can support professional decision-making in a setting where the stakes are high and where trust, justification, and interpretability are essential. The article combines recommendation with large language models to enrich representations and generate explanations, illustrating how technically sophisticated systems may still be oriented toward a support function grounded in clinical practice.

Tolomei et al. [15] bring a similar logic to the domain of education. Their article focuses on sequential course recommendation as a way of supporting students' academic trajectories and, more broadly, their likelihood of persistence and success. This contribution is relevant because it foregrounds a domain in which recommendation may influence long-term life opportunities rather than short-term consumption choices.

The remaining two papers in this cluster extend the constructive agenda to the challenge of sustainability. Merinov and Ricci [11] study how recommendation salience may influence tourist behavior and experience, with particular attention to the possibility of encouraging less crowded and more sustainable choices. Their article is important because it recasts recommendation not simply as a mechanism for matching existing preferences, but as an intervention that can shape behavior in ways that matter for both individual experience and collective urban sustainability.

Belliato et al. [3] further develop this theme by proposing a personalized framework for sustainability goal assignment. Their article demonstrates how adaptive recommendation can be used to encourage engagement with sustainability-oriented actions and missions over time. As with the tourism article, the contribution is significant not only because of its application domain, but also because it presents recommendation as a mechanism for fostering behavioral change aligned with broader collective goals.

Taken together, the papers in this final cluster show that RS4Good also involves the proactive design of recommender systems that support better decisions, more equitable access to opportunities, and more sustainable patterns of action over time.

6 Conclusion

The papers gathered in this special issue collectively suggest that recommender systems for good should be understood not as a narrow thematic niche, but as a broader reorientation of the recommender systems field. As the opening contributions make clear, this reorientation begins by shifting attention from performance-centered optimization toward questions of societal impact, stakeholder participation, and meaningful evaluation. It then extends into a responsibility-centered agenda that requires recommender systems to be examined through the lenses of risks, justice, accountability, auditing, and explainability. The issue further demonstrates that these concerns are especially pressing in contexts where recommender systems shape the experiences of vulnerable users or mediate access to socially consequential information. Finally, the papers show that recommender systems can also serve as constructive interventions in domains such as healthcare, education, news, tourism, and sustainability.

Taken together, the contributions illustrate both the breadth and the coherence of the emerging RS4Good agenda. They span different domains, technical approaches, and methodological traditions, yet converge on a shared understanding: recommender systems must be evaluated in relation to the social worlds in which they are embedded and which they may in turn influence. Rather than treating social good as an external add-on to recommender systems research, the papers in this special issue suggest that it should increasingly become part of the field's core vocabulary and practice.

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